Appendix A

Data points used in the calibration of the corrosion loss equation. The following bulleted list outlines the characteristics of the data used to calibrate the equation.

- For Steel only 10% of oxidation byproduct remains on metal when inspected
- Low Alloy Steels - Rust is darker in color and finer in grain than formed on ordinary steel
- When bacteria are present corrosion could be as high as 10 mm/year
- Atmospheric/Sea water Corrosion Usually is highest around Low Tide Line
- As one goes deeper and deeper in water the corrosion rate decreases

8 yea

Type of Metal	Remarks	Location	Corrosion Loss (mm)	Corrosion Loss (mils)	Corrosion Rate (mm/yr)	Corrosion Rate (mils/yr)	Corresponding P	Corresponding N
Iron / Wrought Iron								
8 years	Atmosphere	Cristobal, Panama Canal Zone	0.559	22.000	—	—	9.7	—
8 years	In Pacific Ocean	Off Panama Canal	0.406	16.000	—	—	7.75	—
8 years	In Pacific Ocean	Off Panama Canal	—	—	—	—	—	—
8 years	In Pacific Ocean	Off Panama Canal	—	—	—	—	—	—
1 year	Ingot Iron Exposed for a Year (Salt Content of Air <0.2	Nigeria	—	—	0.044	1.73	5	5.1
Mild Steel / Carbon Steel / Cast Iron	Rural or Suburban	Godalming	—	—	0.048	1.890	5	6
1 year		Llanwrtyd Wells	0.200	7.874	0.069	2.717	7,8	6, 4.25

1 year		Teddington	____	____	0.07	2.756	7	6
1 year	Marine	Brixham	____	____	0.053	2.087	5, 6, 7	7, 4, 1.2
Great Britain		Calshot	____	____	0.079	3.110	10, 7.4	1.3, 30
1 year	Industrial	Motherwell	____	____	0.095	3.740	11, 10, 9	1, 5.5, 11
1 year		Woolwich	____	____	0.102	4.016	12, 10, 11	1, 7, 5
1 year		Sheffield	0.750	29.528	0.135	5.315	15, 14, 13	1, 5.75, 8.75
1 year		Frodingha m	____	____	0.16	6.299	15, 16, 17	1, 7, 5.3
1 year		Derby	____	____	0.17	6.693	19, 18, 17	1, 5, 5.7
1 year	Rural or Suburban	Khartoum	____	____	0.003	0.118	0.2, 0.5	1, 2
1 year		Abisko, North Sweden	____	____	0.005	0.197	0.2, 0.5	10, 0.23
1 year		Delhi	____	____	0.008	0.315	0.3, 0.5	0.25, 10
1 year		Basrah	____	____	0.015	0.591	0.5, 1	0.25, 0.7
1 year		State College, PA, USA	____	____	0.043	1.693	5, 3, 6	6, 0.67, 3
1 year		Berlin-Dahlem	____	____	0.053	2.087	5, 6, 7	7, 4, 1.2
1 year	Marine	Singapore	____	____	0.015	0.591	0.5, 1	0.25, 0.7
1 year		Apapa, Nigeria	____	____	0.028	1.102	2.9, 5	1, 1.2
1 year		Sandy Hook, NJ, USA	____	____	0.084	3.307	9.5, 12, 15	1, 3, 1.7
1 year	Marine / Industrial	Congella, South Africa	____	____	0.114	4.488	12.5, 15, 17	1, 3.2, 2.7
1 year	Industrial	Pittsburgh, Pa, USA	____	____	0.108	4.252	12.5, 15, 17	12.5, 15, 17
1 year	Marine, surf beach	Lagos	____	____	0.615	24.213	____	____
				Sum	2.072	81.575		
				Mean	0.094	3.708		
				Standard Deviation	0.126	4.965		

				COV	1.339	1.339		
0.56	Sea Water	total immersion	____	____	0.143	17.294	____	____
0.56	Sea Water	total immersion	____	____	0.143	17.272	____	____
0.56	Sea Water	total immersion	____	____	0.148	17.284	____	____
0.56	Sea Water	total immersion	____	____	0.143	17.379	____	____
0.56	Sea Water	total immersion	____	____	0.140	17.457	____	____
0.56	Sea Water	total immersion	____	____	0.140	17.517	____	____
0.56	Sea Water	total immersion	____	____	0.136	17.575	____	____
0.56	Sea Water	total immersion	____	____	0.143	17.628	____	____
0.56	Sea Water	total immersion	____	____	0.158	17.672	____	____
				Sum	1.294	17.559		
				Mean	0.144	17.410		
				Standard Deviation	0.006	0.154		
				COV	0.043	0.009		
15 years		Halifax, Nova Scotia	____	____	0.108	4.252	34	1
Mild Steel	Natural Water	Plymouth	____	____	0.065	2.559	20	1
5 years	Natural Water	Emsworth	____	____	0.065	2.559	14, 12	1, 3
15 years	Natural Water	Plymouth (reservoir)	____	____	0.043	1.693	13	____
5 years	Natural Water	La Cadene (granite bed)	____	____	0.068	2.677	14, 12	1.3
5 years	Natural Water	Dole (highly calcerous water)	____	____	0.010	0.394	2	1.25
8 years	Natural Water	Rotherham	1.000	39.370	____	____	19.3	____
15 years	Marine Atmosphe	Colombo, Ceylon	____	____	____	____	____	____

	re							
15 years	Marine Atmosphere	Auckland, New Zealand	2.430	95.669	____	___	31.5	____
15 years	Marine Atmosphere	Halifax, Nova Scotia	1.640	64.567	____	___	21.2	____
15 years	Marine Atmosphere	Plymouth , New England	1.090	42.913	____	___	14.2	____
15 years	Immersion in Sea Water	Colombo, Ceylon	2.550	100.394	____	___	33	____
15 years	Immersion in Sea Water	Auckland, New Zealand	0.036	1.417	____	___	0.5	____
15 years	Immersion in Sea Water	Halifax, Nova Scotia	2.150	84.646	____	___	27.7	____
15 years	Immersion in Sea Water	Plymouth , New England	1.580	62.205	____	___	20.5	____
15 years	In Sea Water	Colombo, Ceylon	6.500	255.906	____	___	84	____
15 years	In Sea Water	Auckland, New Zealand	2.590	101.969	____	___	33.5	____
15 years	In Sea Water	Halifax, Nova Scotia	1.230	48.425	____	___	15.9	____
15 years	In Sea Water	Plymouth , New England	2.750	108.268	____	___	35.7	____
15 years	Fresh Water		2.200	86.614	____	___	28.5	____
8 years	Atmosphere	Cristobal, Panama Canal Zone	0.254	10.000	____	___	4.9	____
8 years	In Pacific Ocean	Off Panama Canal	1.0795	42.500	____	___	20.6	____
8 years	In Pacific Ocean	Off Panama Canal	1.5748	62.000	____	___	30	____
3.3 years	In Sea Water	Harbor Island, NC	____	____	0.053	2.100	13.5, 10, 15	1, 1.33, 0.9

7.5 years	In Sea Water	Kure Beach, NC	____	____	0.102	4.000	23.7	1
8 years	In Sea Water	Kure Beach, NC	____	____	0.056	2.200	13	1
23.6 years	In Sea Water	Santa Barbara, CA	____	____	0.038	1.500	14	1
16 years	In Sea Water	Panama Canal (Pac. O.)	____	____	0.069	2.700	23	1
1.5 years	In Sea Water	San Diego (Polluted Sea-water	____	____	0.056	2.200	10, 6	1, 12
5 years	Immersed in Sea Water	Auckland, New Zealand	2.223	87.500	____	____	57	____
5 years	Immersed in Sea Water	Halifax, Nova Scotia	1.039	40.900	____	____	26.5	____
5 years	Immersed in Sea Water	Plymouth, New England	1.717	67.600	____	____	43.6	____
5 years	Immersed in Sea Water	Colombo, Ceylon	3.747	147.500	____	____	95	1
10 years	Outdoor	Sheffield	0.400	15.748	____	____	6.65	____
Low Alloy Steels		1st and 2nd years	____	____	0.077	3.031	23.5, 15, 10	1, 1.6, 4
		6th to 15th year	____	____	0.025	0.984	6.6	1
8 years	Atmosphe re	Cristobal, Panama Canal Zone	0.1905	7.5	____	____	3.65	1
8 years	In Pacific Ocean	Off Panama Canal	____	____	____	____	____	____
10 years	Outdoor	Sheffield	0.175	6.890	____	____	2.9	____
8 years		Rotherham	0.210	8.268	____	____	4	____
Stainless Steels	0.31	Heavy Industrial site	0.081	3.189	____	____	0.94	____
18 years	1.44	Heavy Industrial site	0.052	2.047	____	____	0.6	____

Atmospheric Exposure Tests	2.7	Heavy Industrial site	0.036	1.398	—	—	0.41	—
18 years	3.45	Heavy Industrial site	0.018	0.689	—	—	0.2	—
18 years	304S15	Rural	0.020	0.787	—	—	0.23	—
18 years	304S15	Semi-industrial	0.021	0.827	—	—	0.24	—
18 years	304S15	Heavy Industrial site	0.081	3.189	—	—	0.94	—
18 years	304S15	Marine	0.085	3.346	—	—	1	—
18 years	316S33	Rural	0.018	0.689	—	—	0.2	—
18 years	316S33	Semi-industrial	0.018	0.709	—	—	0.21	—
18 years	316S33	Heavy Industrial site	0.036	1.398	—	—	0.41	—
18 years	316S33	Marine	0.024	0.945	—	—	0.28	—
Maraging Steels		244m from the sea	—	—	0.005	0.197	1.21	1
8 years		in sea water flowing at 0.6 m/s	—	—	0.05	1.969	12, 21	1, 0.4
Nickel Iron Alloys	Fe36Ni	Colombo, Ceylon	0.000		—	—	—	—
Marine Atmosphere	Fe36Ni	Auckland, New Zealand	0.000	0.000	—	—	—	—
15 years	Fe36Ni	Halifax, Nova Scotia	0.100	3.937	—	—	1.3	—
15 years	Fe36Ni	Plymouth , New England	0.190	7.480	—	—	2.45	—
Immersion in Sea water	Fe36Ni	Colombo, Ceylon	1.000	39.370	—	—	12.9	—
15 years	Fe36Ni	Auckland, New Zealand	0.240	9.449	—	—	3.1	—
15 years	Fe36Ni	Halifax, Nova Scotia	2.590	101.969	—	—	33.5	—

15 years	Fe36Ni	Plymouth , New England	0.250	9.843	___	___	3.24	___
In Sea Water	Fe36Ni	Colombo, Ceylon	2.500	98.425	___	___	32.4	___
15 years	Fe36Ni	Auckland, New Zealand	1.080	42.520	___	___	14	___
15 years	Fe36Ni	Halifax, Nova Scotia	3.490	137.402	___	___	45	___
15 years	Fe36Ni	Plymouth , New England	1.820	71.654	___	___	23.5	___
Fresh Water (15 yrs)	Fe36Ni		2.000	78.740	___	___	25.8	___
Copper Steel	Atmosphere (8 years)	Cristobal, Panama Canal Zone	0.432	17.000	___	___	5.6	___

Appendix B

Example of several hypothetical distributions that were calculated for flaw sizes ranging from ¼ inches to 8 inches. The ranges for which calculations were performed were for ¼ inch flaws, 1 inch flaws, 2 inch flaws, 5 inch flaws and 8 inch flaws.

1/4 inch Flaws

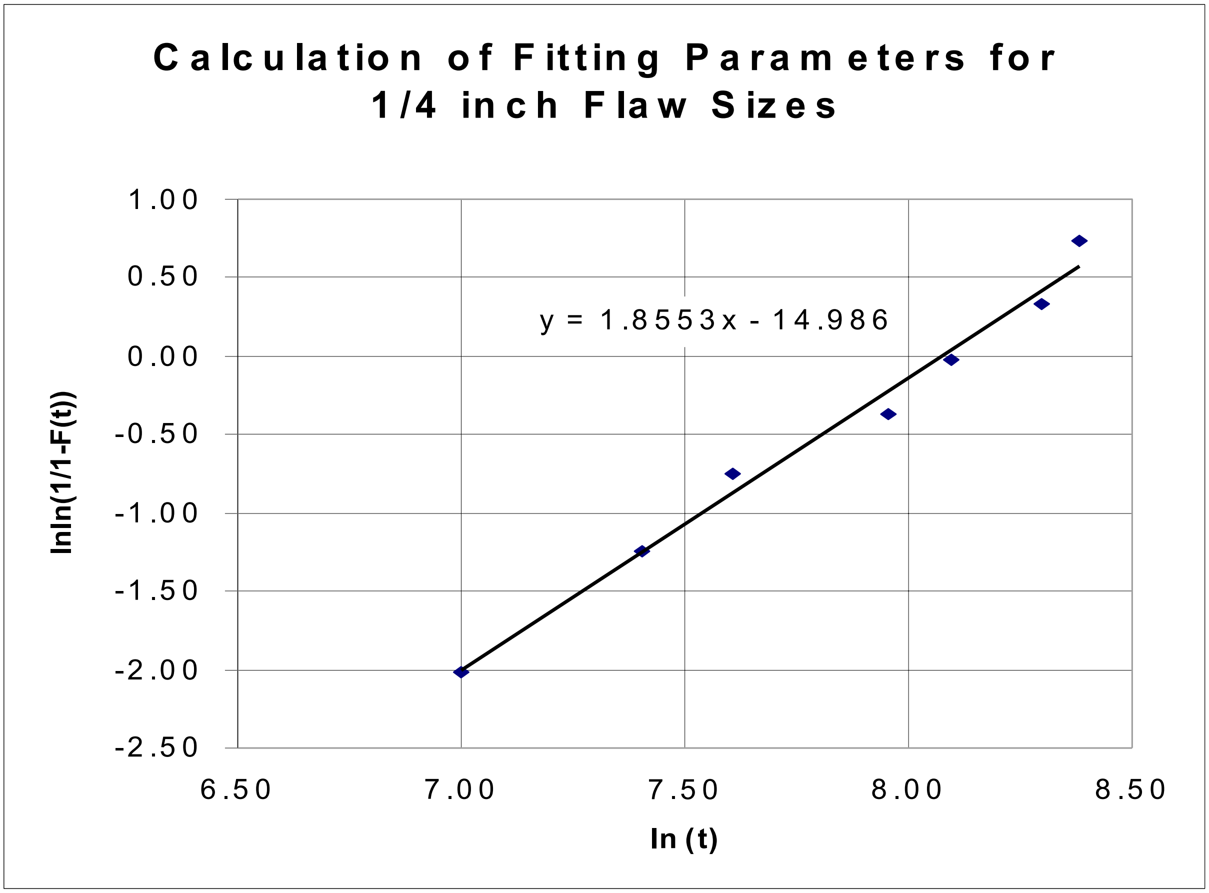
Time to develop 300, 0.25 inch flaw sizes over a length of 1 mile.

Observation Number	Time (years)	Time (days)	ln(t)	lnln(1/1-F(t))
1	3	1095	7.00	-2.01
2	4.5	1642.5	7.40	-1.25
3	5.5	2007.5	7.60	-0.76
4	7.8	2847	7.95	-0.37
5	9	3285	8.10	-0.02
6	11	4015	8.30	0.33
7	12	4380	8.38	0.73
8	13.1	4781.5	8.47	

Slope	Intercept
1.855	-14.986

κ	λ
1.855	3.10E-04

Time (Days)	S(t)	f(t)	h(t)	H(t)
500	0.969	1.13E-04	1.17E-04	0.032
1000	0.892	1.89E-04	2.12E-04	0.114
1500	0.785	2.35E-04	3.00E-04	0.242
2000	0.662	2.54E-04	3.83E-04	0.413
2500	0.535	2.48E-04	4.64E-04	0.625
3000	0.416	2.26E-04	5.42E-04	0.877
3500	0.311	1.93E-04	6.18E-04	1.167
4000	0.224	1.56E-04	6.93E-04	1.495
4500	0.156	1.19E-04	7.67E-04	1.860
5000	0.104	8.74E-05	8.39E-04	2.261
5500	0.067	6.13E-05	9.10E-04	2.699
6000	0.042	4.11E-05	9.81E-04	3.172
6500	0.025	2.65E-05	1.05E-03	3.679
7000	0.015	1.64E-05	1.12E-03	4.222
7500	0.008	9.79E-06	1.19E-03	4.798
8000	0.004	5.62E-06	1.25E-03	5.408
8500	0.002	3.11E-06	1.32E-03	6.052
9000	0.001	1.66E-06	1.39E-03	6.729
9500	0.001	8.54E-07	1.45E-03	7.439


Calculation of Fitting Parameters for
1/4 inch Flaw Sizes
y = 1.8553x - 14.986
lnln(1/1-F(t))
ln (t)
1.00
0.50
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
6.50
7.00
7.50
8.00
8.50

1inch Flaws

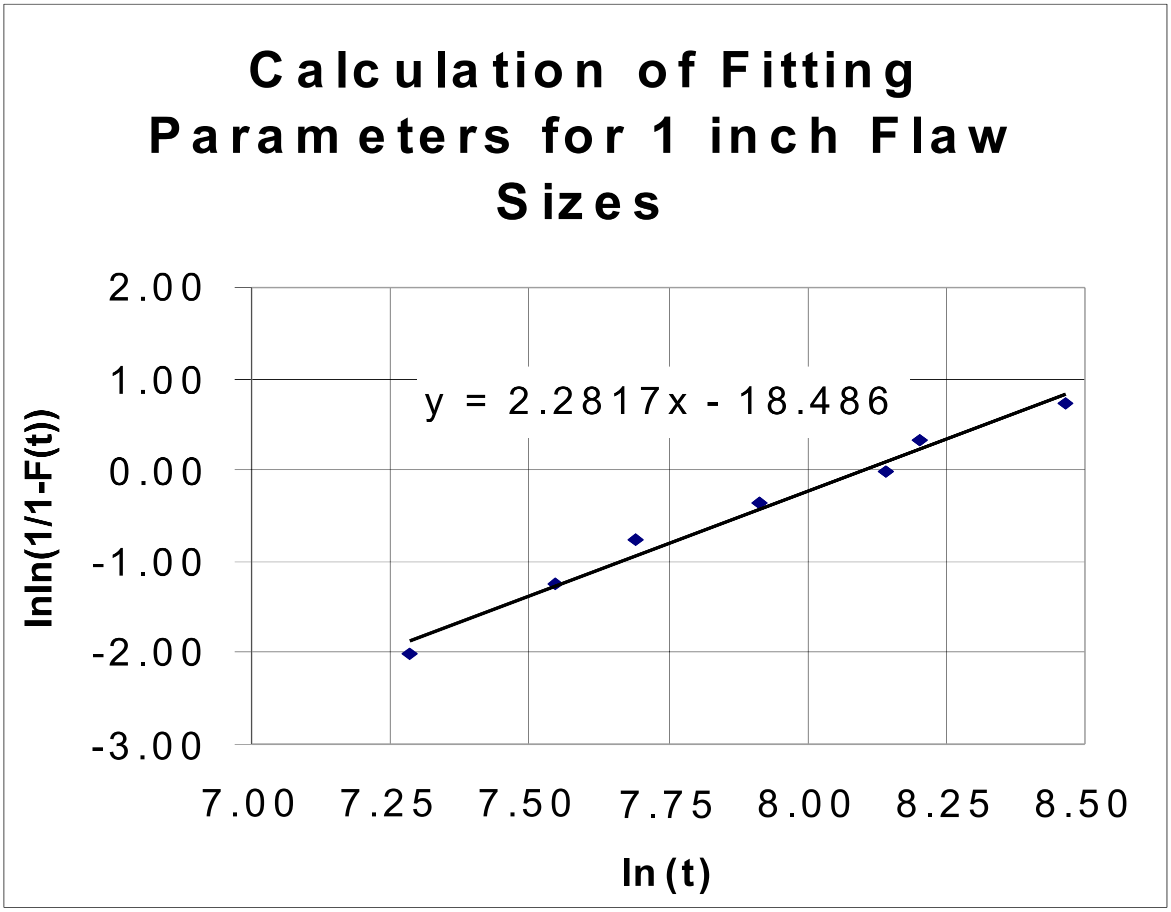
Time to develop 150, 1 inch flaw sizes over a length of 1 mile.

Observation Number	Time (years)	Time (days)	ln(t)	lnln(1/1-F(t))
1	4	1460	7.29	-2.01
2	5.2	1898	7.55	-1.25
3	6	2190	7.69	-0.76
4	7.5	2737.5	7.91	-0.37
5	9.4	3431	8.14	-0.02
6	10	3650	8.20	0.33
7	13	4745	8.46	0.73
8	16	5840	8.67	

Slope	Intercept
2.282	-18.486

κ	λ
2.282	3.03E-04

Time (Days)	S(t)	f(t)	h(t)	H(t)
500	0.987	6.07E-05	6.15E-05	0.013
1000	0.937	1.40E-04	1.50E-04	0.066
1500	0.848	2.13E-04	2.52E-04	0.165
2000	0.727	2.64E-04	3.64E-04	0.319
2500	0.588	2.85E-04	4.84E-04	0.531
3000	0.447	2.74E-04	6.12E-04	0.804
3500	0.319	2.38E-04	7.45E-04	1.143
4000	0.212	1.88E-04	8.84E-04	1.550
4500	0.132	1.35E-04	1.03E-03	2.028
5000	0.076	8.92E-05	1.18E-03	2.580
5500	0.041	5.39E-05	1.33E-03	3.206
6000	0.020	2.98E-05	1.49E-03	3.911
6500	0.009	1.51E-05	1.65E-03	4.694
7000	0.004	6.98E-06	1.81E-03	5.559
7500	0.001	2.96E-06	1.98E-03	6.507
8000	0.001	1.14E-06	2.15E-03	7.539
8500	0.000	4.04E-07	2.32E-03	8.657
9000	0.000	1.30E-07	2.50E-03	9.863
9500	0.000	3.82E-08	2.68E-03	11.158


Calculation of Fitting Parameters for 1 inch Flaw Sizes
y = 2.2817x - 18.486
lnln(1/1-F(t))
2.00
1.00
0.00
-1.00
-2.00
-3.00
7.00
7.25
7.50
7.75
8.00
8.25
8.50
ln (t)

2inch Flaws

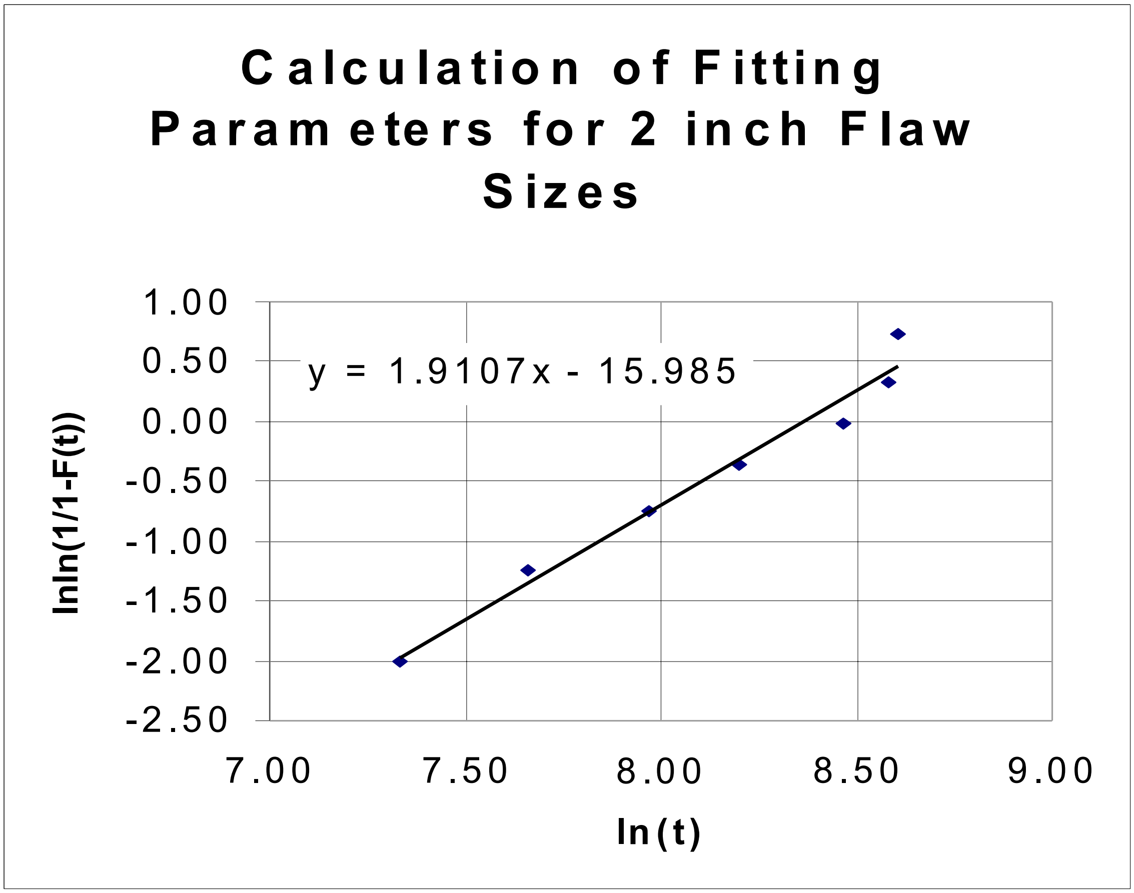
Time to develop 75, 2 inch flaw sizes over a length of 1 mile.

Observation Number	Time (years)	Time (days)	ln(t)	lnln(1/1-F(t))
1	4.2	1533	7.33	-2.01
2	5.8	2117	7.66	-1.25
3	7.9	2883.5	7.97	-0.76
4	10	3650	8.20	-0.37
5	13	4745	8.46	-0.02
6	14.6	5329	8.58	0.33
7	15	5475	8.61	0.73
8	18	6570	8.79	

Slope	Intercept
1.911	-15.985

κ	λ
1.911	2.33E-04

Time (Days)	S(t)	f(t)	h(t)	H(t)
500	0.984	6.16E-05	6.27E-05	0.016
1000	0.940	1.11E-04	1.18E-04	0.062
1500	0.875	1.49E-04	1.70E-04	0.134
2000	0.793	1.76E-04	2.21E-04	0.232
2500	0.701	1.90E-04	2.71E-04	0.355
3000	0.605	1.94E-04	3.20E-04	0.503
3500	0.509	1.88E-04	3.69E-04	0.675
4000	0.418	1.74E-04	4.16E-04	0.871
4500	0.336	1.56E-04	4.63E-04	1.091
5000	0.263	1.34E-04	5.10E-04	1.335
5500	0.202	1.12E-04	5.56E-04	1.601
6000	0.151	9.09E-05	6.02E-04	1.891
6500	0.110	7.15E-05	6.48E-04	2.204
7000	0.079	5.47E-05	6.93E-04	2.539
7500	0.055	4.07E-05	7.38E-04	2.897
8000	0.038	2.95E-05	7.83E-04	3.277
8500	0.025	2.09E-05	8.27E-04	3.679
9000	0.017	1.44E-05	8.71E-04	4.104
9500	0.011	9.67E-06	9.15E-04	4.550


Calculation of Fitting Parameters for 2 inch Flaw Sizes
y = 1.9107x - 15.985
lnln(1/1-F(t))
ln(t)
1.00
0.50
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
7.00
7.50
8.00
8.50
9.00

<u>5 inch Flaws</u>

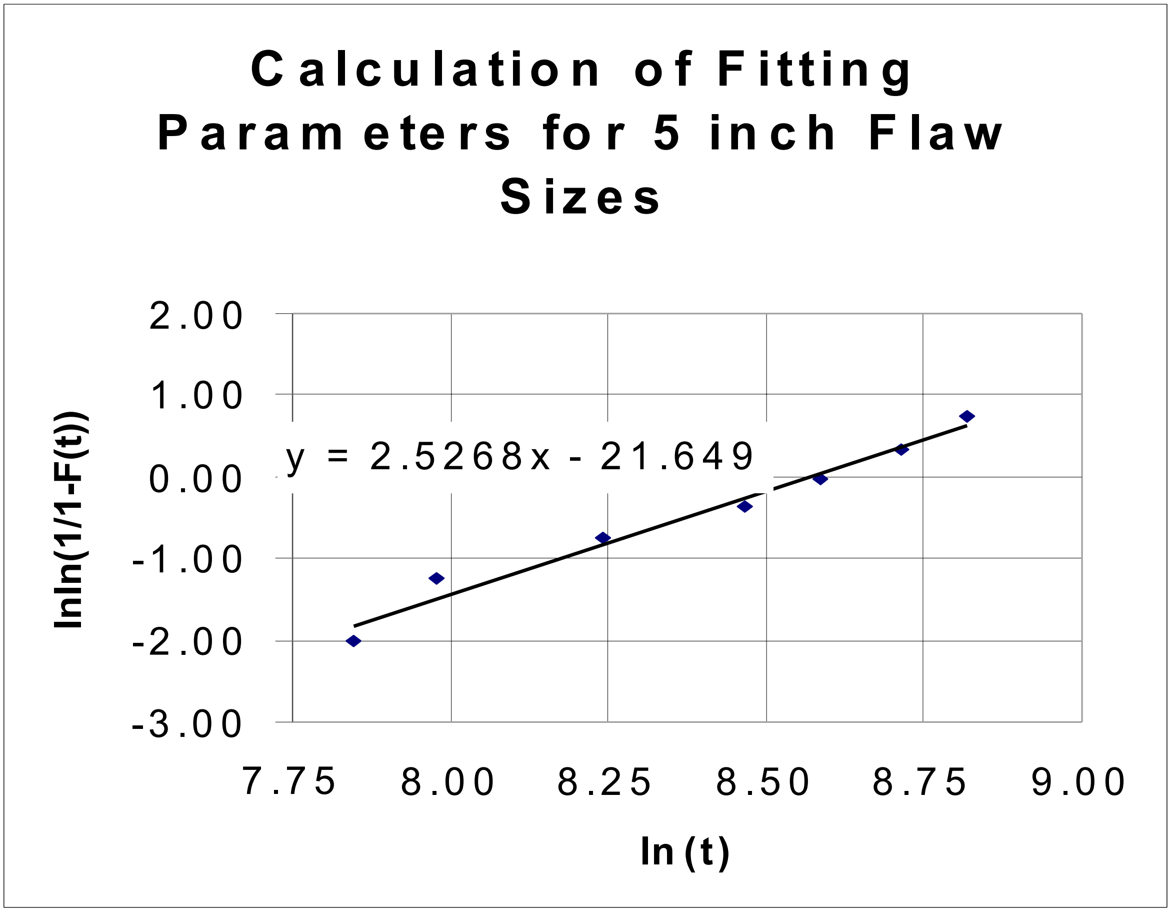
Time to develop 25, 5 inch flaw sizes over a length of 1 mile.

Observation Number	Time (years)	Time (days)	ln(t)	lnln(1/1-F(t))
1	7	2555	7.85	-2.01
2	8	2920	7.98	-1.25
3	10.4	3796	8.24	-0.76
4	13	4745	8.46	-0.37
5	14.7	5365.5	8.59	-0.02
6	16.7	6095.5	8.72	0.33
7	18.5	6752.5	8.82	0.73
8	20	7300	8.90	

Slope	Intercept
2.527	-21.649

κ	λ
2.527	1.90E-04

Time (Days)	S(t)	f(t)	h(t)	H(t)
500	0.997	1.32E-05	1.32E-05	0.003
1000	0.985	3.75E-05	3.81E-05	0.015
1500	0.959	6.78E-05	7.08E-05	0.042
2000	0.917	1.01E-04	1.10E-04	0.087
2500	0.858	1.32E-04	1.54E-04	0.153
3000	0.785	1.60E-04	2.04E-04	0.242
3500	0.700	1.80E-04	2.58E-04	0.357
4000	0.606	1.92E-04	3.16E-04	0.501
4500	0.509	1.93E-04	3.79E-04	0.674
5000	0.415	1.84E-04	4.45E-04	0.880
5500	0.326	1.68E-04	5.14E-04	1.120
6000	0.248	1.46E-04	5.88E-04	1.395
6500	0.181	1.20E-04	6.64E-04	1.708
7000	0.128	9.48E-05	7.43E-04	2.059
7500	0.086	7.12E-05	8.26E-04	2.452
8000	0.056	5.09E-05	9.12E-04	2.886
8500	0.035	3.46E-05	1.00E-03	3.364
9000	0.021	2.24E-05	1.09E-03	3.886
9500	0.012	1.38E-05	1.18E-03	4.455


Calculation of Fitting Parameters for 5 inch Flaw Sizes
y = 2.5268x - 21.649
lnln(1/1-F(t))
2.00
1.00
0.00
-1.00
-2.00
-3.00
7.75
8.00
8.25
8.50
8.75
9.00
ln(t)

<u>8 inch Flaws</u>

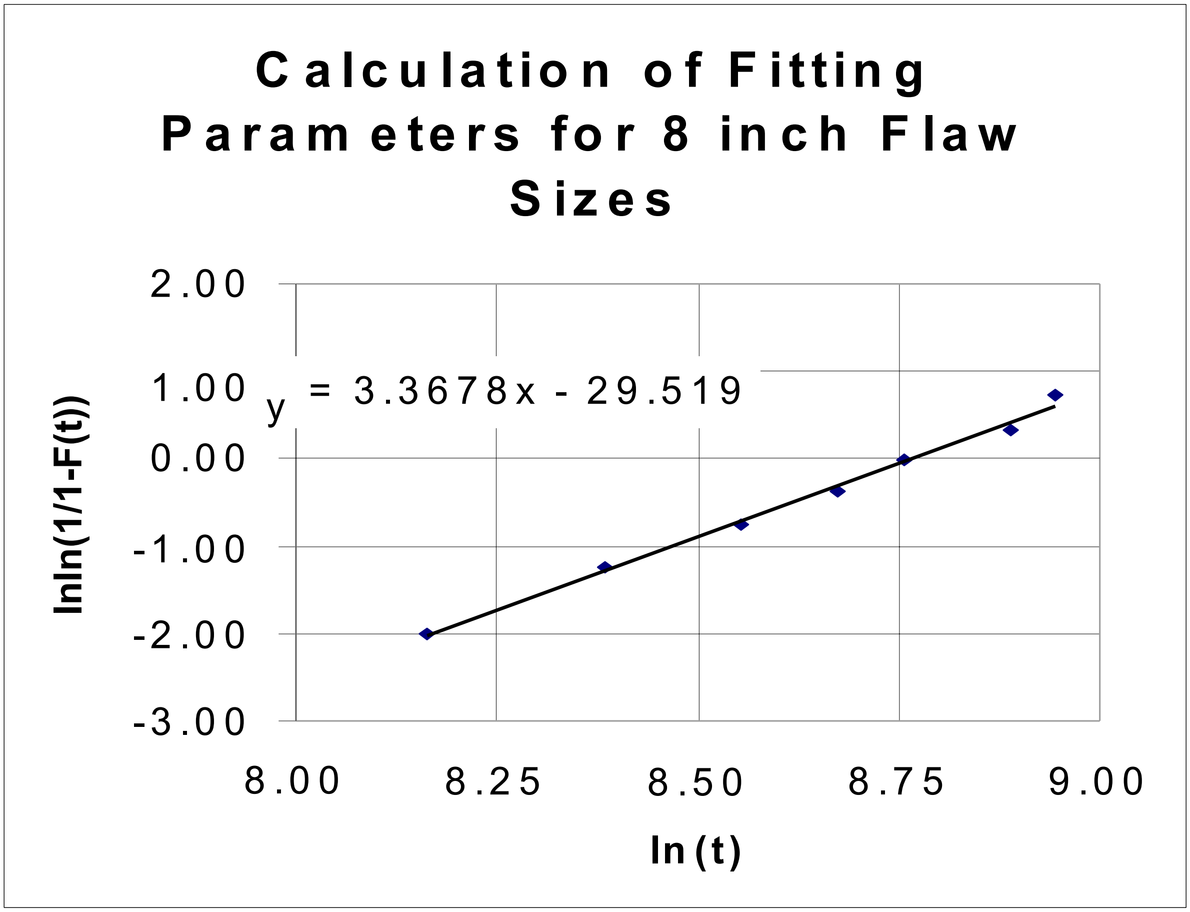
Time to develop 5, 8 inch flaw sizes over a length of 1 mile.

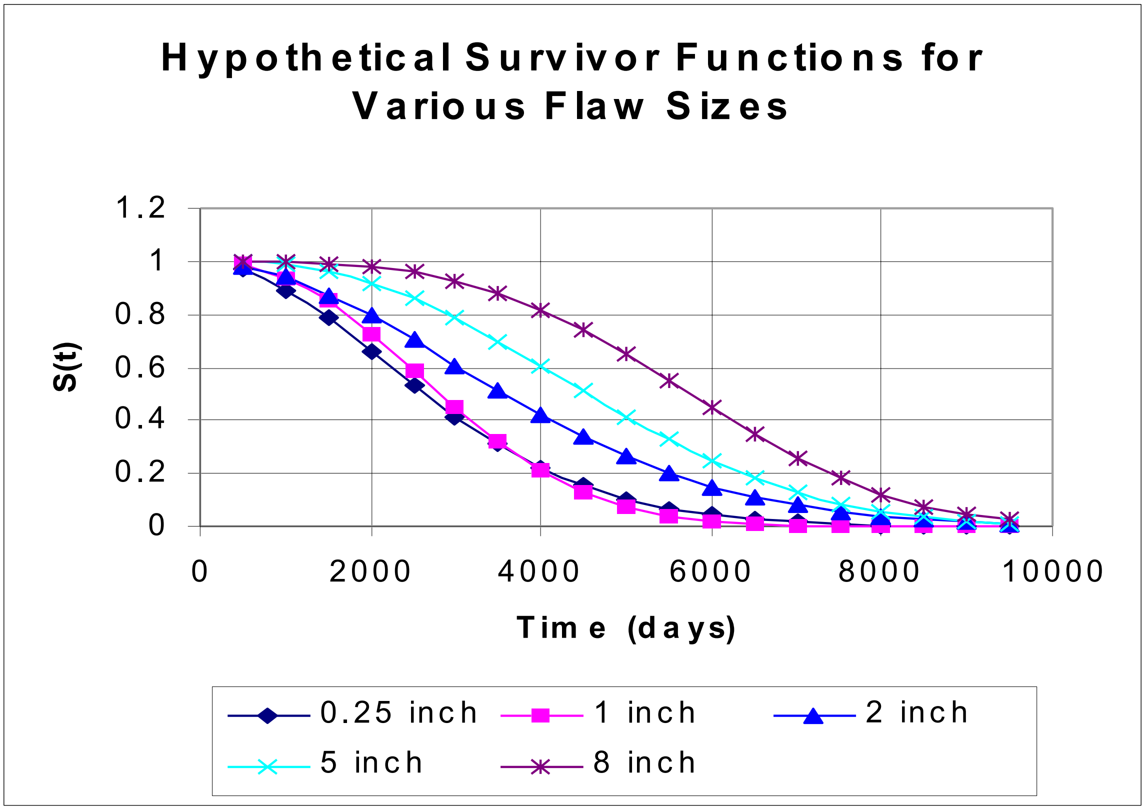
Observation Number	Time (years)	Time (days)	ln(t)	lnln(1/1-F(t))
1	9.6	3504	8.16	-2.01
2	12	4380	8.38	-1.25
3	14.2	5183	8.55	-0.76
4	16	5840	8.67	-0.37
5	17.4	6351	8.76	-0.02
6	19.9	7263.5	8.89	0.33
7	21	7665	8.94	0.73
8	22	8030	8.99	

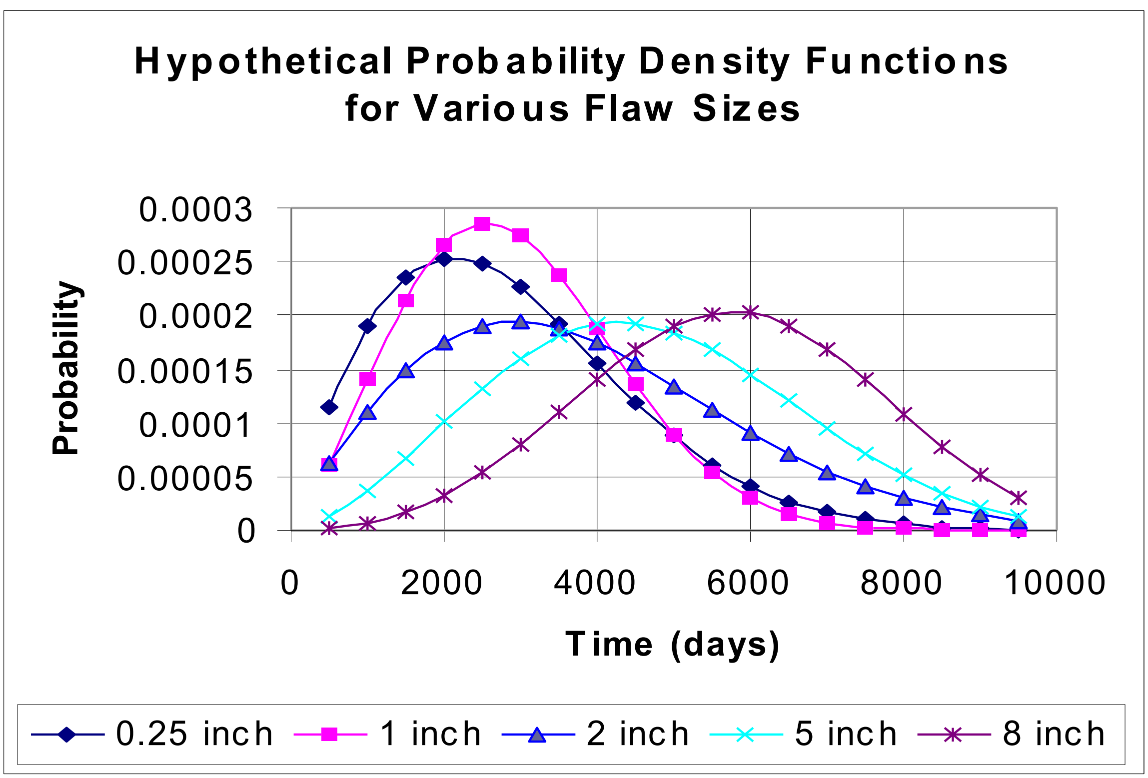
Slope	Intercept
3.368	-29.519

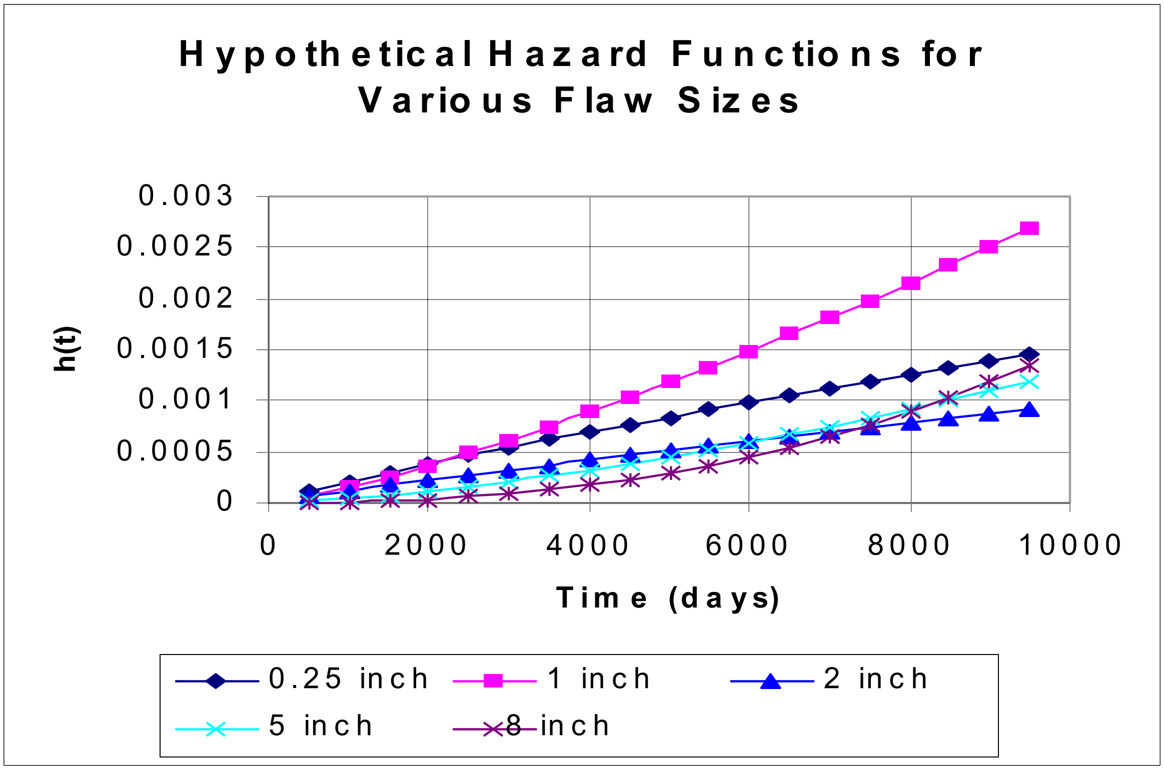
κ	λ
3.368	1.56E-04

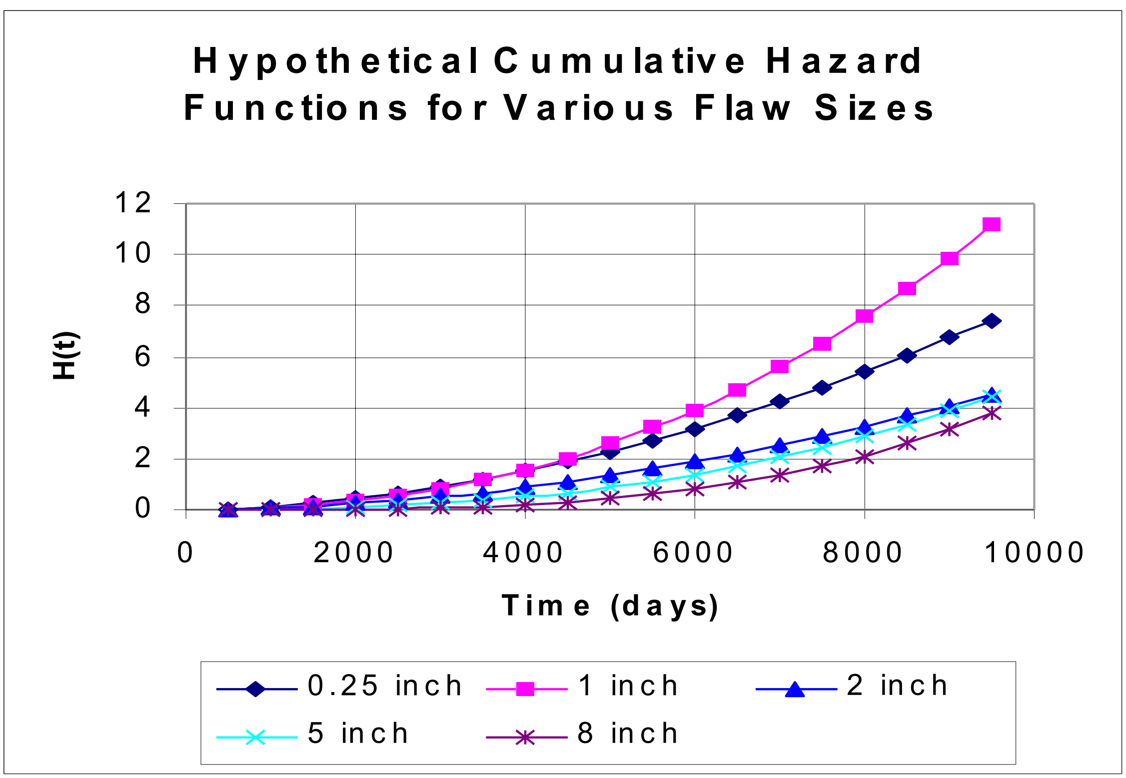
Time (Days)	S(t)	f(t)	h(t)	H(t)
500	1.000	1.25E-06	1.25E-06	0.000
1000	0.998	6.46E-06	6.47E-06	0.002
1500	0.993	1.68E-05	1.69E-05	0.008
2000	0.980	3.27E-05	3.34E-05	0.020
2500	0.959	5.43E-05	5.66E-05	0.042
3000	0.925	8.07E-05	8.72E-05	0.078
3500	0.878	1.10E-04	1.26E-04	0.131
4000	0.815	1.40E-04	1.72E-04	0.205
4500	0.738	1.68E-04	2.28E-04	0.304
5000	0.648	1.89E-04	2.92E-04	0.434
5500	0.550	2.01E-04	3.66E-04	0.598
6000	0.448	2.02E-04	4.50E-04	0.802
6500	0.350	1.90E-04	5.44E-04	1.050
7000	0.260	1.68E-04	6.48E-04	1.348
7500	0.183	1.39E-04	7.63E-04	1.700
8000	0.121	1.08E-04	8.89E-04	2.113
8500	0.075	7.69E-05	1.03E-03	2.592
9000	0.043	5.08E-05	1.18E-03	3.142
9500	0.023	3.08E-05	1.34E-03	3.769


Calculation of Fitting Parameters for 8 inch Flaw Sizes
y = 3.3678x - 29.519
2.00
1.00
0.00
-1.00
-2.00
-3.00
lnln(1/1-F(t))
8.00
8.25
8.50
8.75
9.00
ln(t)


Hypothetical Survivor Functions for Various Flaw Sizes
S(t)
1.2
1
0.8
0.6
0.4
0.2
0
0
2000
4000
6000
8000
10000
Time (days)
0.25 inch
1 inch
2 inch
5 inch
8 inch


Hypothetical Probability Density Functions for Various Flaw Sizes
Probability
0.0003
0.00025
0.0002
0.00015
0.0001
0.00005
0
0
2000
4000
6000
8000
10000
Time (days)
0.25 inch
1 inch
2 inch
5 inch
8 inch


Hypothetical Hazard Functions for Various Flaw Sizes
h(t)
0.003
0.0025
0.002
0.0015
0.001
0.0005
0
0
2000
4000
6000
8000
10000
Time (days)
0.25 inch
1 inch
2 inch
5 inch
8 inch


Hypothetical Cumulative Hazard Functions for Various Flaw Sizes
H(t)
12
10
8
6
4
2
0
0
2000
4000
6000
8000
10000
Time (days)
0.25 inch
1 inch
2 inch
5 inch
8 inch